

Archaeologist

WEIGHT 3 · LONGEST (Nykolas Rekasius) · 17+3

Three-pole map · CLEAR generative-vs-latent labels · inventum · Cosmos family evolution · PARALLEL to JEPA, not child.

ONE-LINER

Open Physical-AI generative-video world-simulator pole — successor to Predict1/Transfer1, PARALLEL to JEPA latent WMs, not their child.

SPINE: Generative video as physics simulator — Predict2.5 (flow T2W/I2W/V2W) + Reason1 + Transfer2.5 | PARALLEL to latent WMs (Dreamer/JEPA/LeWM)
| synth data · policy eval · closed-loop robotics/AV

Talk arc — densest lineage pack (Nykolas)

Priors → inventum → three poles → family → foils → apps

01-03

Spine + three-pole map

A discriminative | B predictive JEPA→LeWM | C generative Cosmos (THIS)

04-06

CLEAR priors + inventum

Predict1/FM/DiT/Reason1/Dreamer foil · V-JEPA 2 FOIL not ancestor

07-10

Family + architecture figs

Predict1→2.5→Transfer2.5 · Fig.2 arch · Fig.6-7 prefs

11-14

Apps evidence

Fig.9-10 Transfer · Fig.11-13 robot 24/30 · Fig.14-15 AV · Fig.20 Bridge

15-17

Labels + tattoo + close

CLEAR generative-vs-latent on EVERY claim · do not fold into JEPA

Three-pole teaching map — Dec 7 fork

Do NOT collapse Pole C into Pole B

Pole A

Discriminative

SigLIP / DINO →
VLA perception towers
($\pi_0...$ / DINO-WM freeze)

PERCEPTION

Pole B

Predictive latent

I/V-JEPA → LeWM / 2-AC
LATENT plan · fast CEM
cheap imagination

LATENT WM

Pole C ★

Generative video

Sora-class → Cosmos
Predict/Transfer (THIS)
pixel simulator · synth/eval

GENERATIVE WFM

Syllabus moral after LeWM

B = fast trainable imagination/planning · C = rich observation synthesis + visual domain randomization. Same “world model” phrase, opposite compute contracts.

CLEAR priors — cited / direct ancestry

Tag every node GENERATIVE vs LATENT

Cosmos-Predict1 / Transfer1	Immediate WFM predecessor (EDM DiT; T5; 7B Transfer)	GENERATIVE
Flow matching (Lipman'22)	Training objective replacing Predict1 EDM	GENERATIVE
DiT + Wan2.1 VAE	Backbone + tokenizer (4×8×8)	GENERATIVE
Cosmos-Reason1	Physical AI VLM text encoder (replaces T5)	GENERATIVE
Ha & Schmidhuber / Dreamer	Latent WM paradigm FOIL (§7)	LATENT FOIL
V-JEPA 2 Assran'25	Cited latent predictive WM example — FOIL not ancestor	LATENT FOIL
ControlNet-style / Transfer1	Transfer2.5 parent design	GENERATIVE
Diffusion Policy · DreamGen	Policy / synth-data consumers	APPS
Bridge / DROID / AgiBot / GR00T	Robotics data priors (cited)	DATA

Do NOT fold — TENTATIVE / false equivalences

Police the JEPA-category error

FALSE

Cosmos as “next I-JEPA”

Different objective: generate pixels vs predict embeddings

FALSE

Transfer Real2Real = V-JEPA 2-AC Franka CEM

Data aug $\neq$ latent CEM planning

FALSE

π_0 / OpenVLA as paper-cited consumers

Not cited — rhyme $\neq$ citation

TENTATIVE

Pixel-space CEM with Predict2.5 as dynamics

Research fork; compute-hostile vs LeWM

TENTATIVE

BEHAVIOR-1K SR from Cosmos-aug VLAs

Plausible; not measured here

TENTATIVE

Distilled 4-step as real-time closed-loop renderer

PAI near-parity; robot Hz not reported

CLEAR inventum — what THIS paper adds

Unified FM Predict2.5 · 200M · Transfer 2B>7B · apps suite

Unified FM Predict2.5

Flow-matching DiT 2B/14B; T2W/I2W/V2W via frame-replace; Reason1 text.

200M curated clips

From 35M h raw (~4% survive);
Physical-AI domain packs + progressive
256→720p.

Post-train stack

Domain SFT → model soup → VideoAlign
RL → optional rCM 4-step distill.

Transfer2.5 2B > 7B

3.5× smaller than Transfer1; better
adherence + long-horizon RNDS.

Apps suite

Robot 24/30 · AV LET-AP 0.394 ·
DreamGen GR1 · Bridge AC 24.95.

Open release

nvidia-cosmos/predict2.5 + transfer2.5
· NVIDIA Open Model License.

Cosmos family evolution (GENERATIVE lineage)

Predict1 → Predict2.5 + Reason1 → Transfer2.5 — CLEAR succession

Predict1

EDM diffusion DiT + T5
separate mode recipes



Predict2.5

Flow matching · unify T2W/I2W/V2W
Reason1 · SFT soup · RL · distill



Transfer2.5

Control-net on 2B base
every-7 blocks · 3.5× smaller

auto/multiview

7-cam driving

robot/action-cond

Bridge AC

robot/multiview-agibot

gripper synth

gr00tdream-gr1 14B

DreamGen

Transfer edge/blur/depth/seg

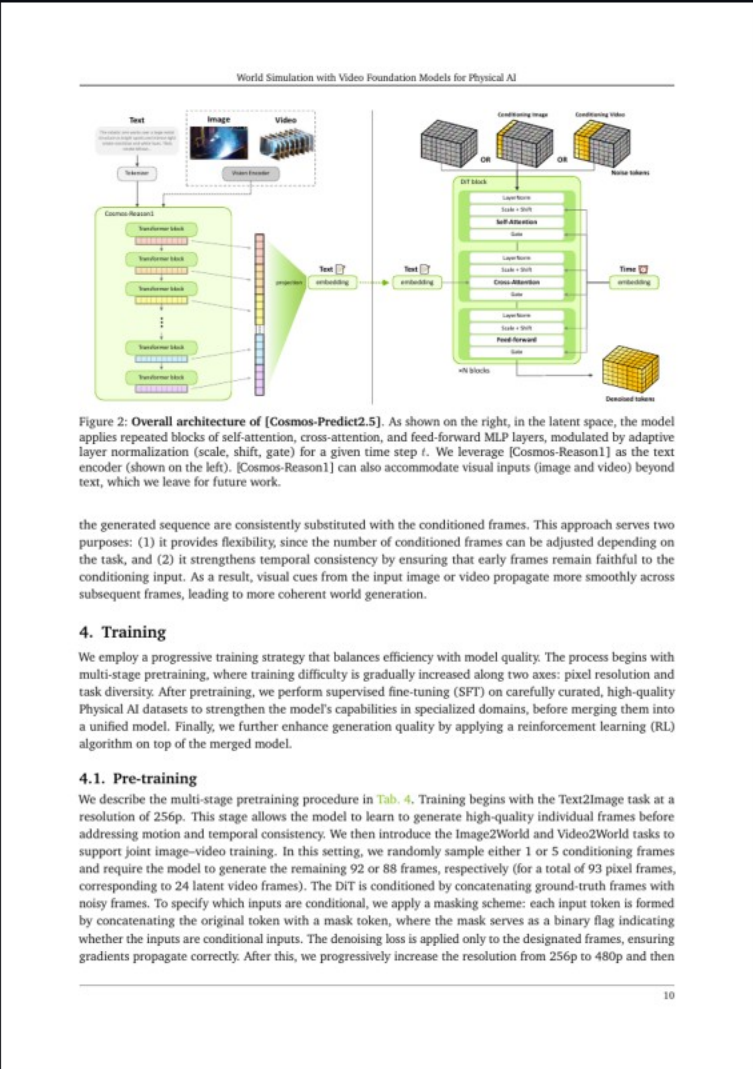
Sim2Real

Transfer auto/robot mv

map / wrist

Fig. 2 — Predict2.5 architecture (CLEAR inventum)

GENERATIVE: Reason1 text + DiT AdaLN · frame-replace I/V conditioning · Wan VAE



Paper Fig. 2 — velocity prediction DiT; Cosmos-Reason1 multi-block → 1024-d cross-attn; 93 frames @16 fps.

Two paradigms (§7) — foil card

CLEAR labels on every row — PARALLEL poles

Axis	LATENT WM	GENERATIVE WFM (THIS)
State	Compact emb / RSSM / CLS	VAE latent → decode pixels
Objective	Predict next repr (+ anti-collapse)	Flow/diffusion match video dist.
Control use	Fast MPC/CEM in latent	AC video; policy eval / synth demos
Synth data	Weak for appearance diversity	STRENGTH — Transfer / DreamGen
Course smoke	Trainable on 1 GPU (siblings)	Infer-only probes (this pack)
Failure	Collapse / poor visuals	Drift / cost / physics imperfect

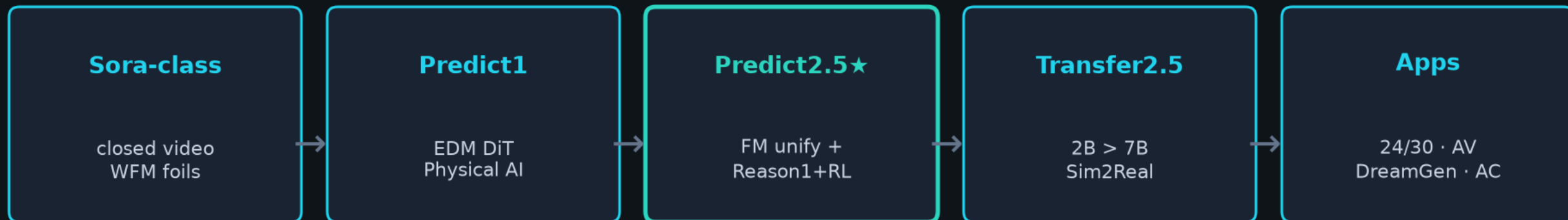
Genealogy — LATENT fork vs GENERATIVE fork

Ha/Dreamer → I-JEPA → V-JEPA → LeWM || Predict1 → Predict2.5/Transfer2.5

LATENT WM



GENERATIVE WFM



PARALLEL — not JEPA child. Paper §7 cites V-JEPA 2 as latent-paradigm example; Cosmos ADOPTS the generative pole.

Fig. 6-7 — human prefs vs Wan (GENERATIVE foil)

2B preferred over Wan2.2-5B; 14B 48.6% vs Wan2.1-14B 31.8%

World Simulation with Video Foundation Models for Physical AI

Figure 6: Despite being of smaller size, post-trained [Cosmos-Predict2.5-2B] is more preferred by humans over Wan2.2 5B and is on par with Wan2.1 14B across a diverse set of prompts.

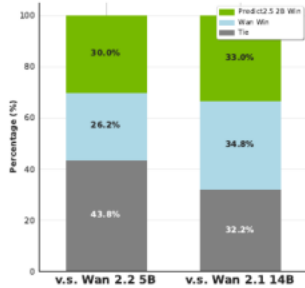
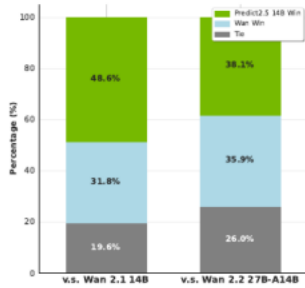


Figure 7: Across a diverse set of prompts, post-trained [Cosmos-Predict2.5-14B] is preferred more often than Wan 2.1 14B, and achieves on par performance to Wan 2.2 27B-A14B, despite having only half the parameter count.



6. Applications

We demonstrate the versatility of [Cosmos-Predict2.5] across multiple Physical AI applications. First, we introduce [Cosmos-Transfer2.5], which provides control-net style generation capability to Physical AI applications (Sec. 6.1). Compared to [Cosmos-Transfer1], the new model is substantially more effective while being 3.5× smaller. We further show that [Cosmos-Transfer2.5] enables Real2Real augmentation for policy learning (Sec. 6.2), and that the same paradigm applies to autonomous driving, where we construct multiview world models conditioned on world scenario maps for realistic driving simulation (Sec. 6.3).

World Simulation with Video Foundation Models for Physical AI

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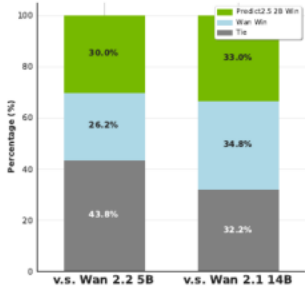
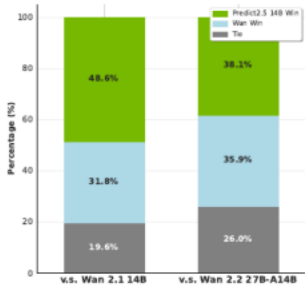


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Fig. 9-10 — Transfer2.5 vs Transfer1

GENERATIVE control-net: 3.5× smaller · better RNDs long-horizon



Fig. 9 visual comparison

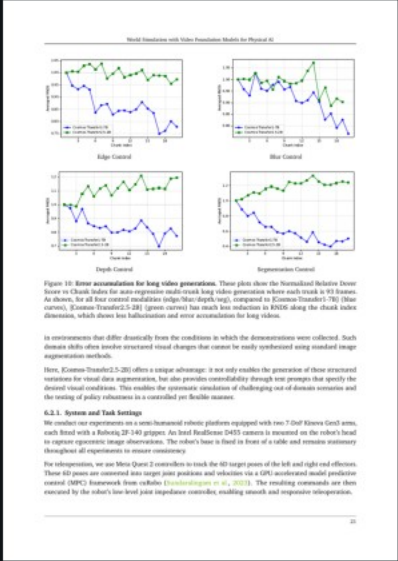


Fig. 10 RNDs curves (less drop)

Family ablate (multi-factor)

Transfer1-7B→Transfer2.5-2B confounds: stronger Predict2.5 base + Physics-AI data + every-7 block placement.
Cite Tab.12 Edge F1 0.49 vs 0.38; Uniform quality 9.31 vs 9.24.

Fig. 11-13 — Real2Real robot 24/30 (CLEAR)

Offline Transfer visual aug → Diffusion Policy — NOT latent CEM



Fig.11 teleop



Fig.12 Transfer augs



Fig.13 rollouts 24/30

1/30

Base 100 demos

5/30

Classic image augs

24/30

Transfer2.5 augs ★

LABEL: synth-data augmentation for BC — different mechanism from V-JEPA 2-AC zero-shot Franka planning. Do NOT equate.

Fig. 14-15 AV + Fig. 20 Bridge AC

GENERATIVE apps: map-conditioned multiview · action-cond policy-eval WM

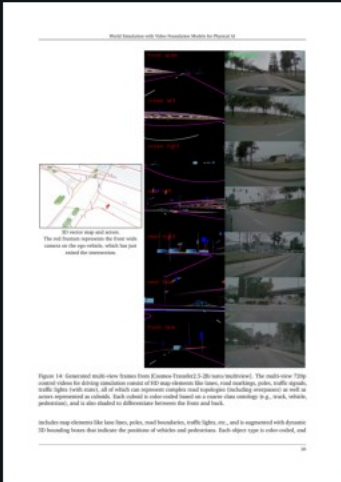


Fig.14 AV multiview

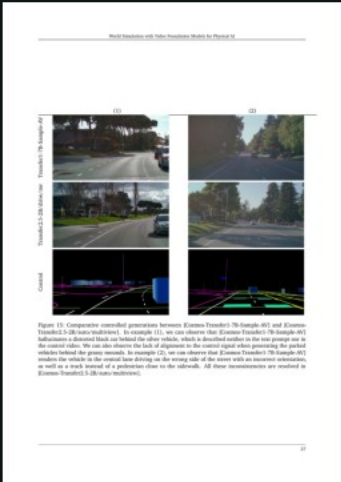


Fig.15 Transfer1 vs 2.5

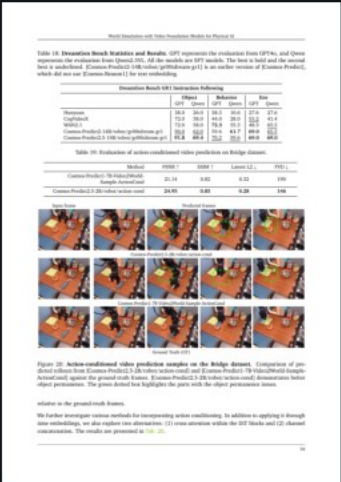


Fig.20 Bridge AC

0.394

LET-AP vs T1 0.243

FVD 23

vs Predict1 AV 63

24.95

Bridge PSNR (TimeEmb)

TimeEmb >>

Cross 24.41 > Concat 23.11

CLEAR vs TENTATIVE — Archaeologist discipline

Preserve paper numbers · generative-vs-latent on every lineage claim

CLEAR

- Predict1 → Predict2.5 successor (GENERATIVE)
- Flow DiT + Reason1 + open repos
- PAI 0.768 T2W / 0.810 I2W
- Robot Transfer aug 24/30 vs 5/30
- Bridge AC Tab.19–20; DreamGen Tab.18 IF
- V-JEPA 2 cited as LATENT FOIL (§7)
- Prior JEPA/LeWM labels UNCHANGED
- GR00T/AgiBot/Bridge/DROID cited data

TENTATIVE / refuse

- Online CEM-in-pixels autonomy SR
- DreamGen → hardware VLA task SR
- Hybrid $\pi_{0.5}$ +LeWM+Cosmos stack (Visionary)
- π_0 /OpenVLA paper citations (absent)
- Newton-solver physics replacement
- Equating Transfer aug with 2-AC CEM
- Folding Cosmos into JEPA genealogy
- Course students train 4096-H100 WFM

Numbers tattoo — pin these (KanBot must-cites)

Shared spine across all five seats · paper numbers only

24/30

Real2Real robot vs 5/30

0.768

PAI T2W Overall post

0.810

PAI I2W Overall post

24.95

Bridge AC PSNR

IF $\neq$ SR

DreamGen Tab.18 $\neq$ VLA

0.394

AV LET-AP Transfer2.5

3.5×

Transfer smaller vs T1

200M

curated video clips

2B / 14B

Predict scales

93f @16fps

~5.8 s generation

PARALLEL

not JEPA child

GENERATIVE

pixel WFM pole

Archaeologist — close

Takeaways + exam bite

1. Pole C generative video WFM sits PARALLEL to Pole B JEPA latent planners — teach the fork.
2. Inventum = unified FM Predict2.5 + Reason1 + Transfer2.5 2B>7B + Physical-AI apps suite.
3. 24/30 is offline synth-data aug; Bridge AC is video metrics for policy eval — not Franka CEM.
4. One-liner: open Physical-AI generative-video world-simulator — not a JEPA child.

EXAM BITE

Name the three poles (discriminative / predictive-latent / generative-WFM) and give one CLEAR Cosmos inventum that is NOT a JEPA descendant claim.